

Alberto Brambila Solórzano, PhD

CFD and Thermal Simulation Engineer

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Engineering portfolio: albertobraso7.github.io

PROFILE

I am a CFD engineer working across thermal management, rotating machinery and multiphase systems, with thirty documented industrial projects. My work connects physical modelling to design decisions: identifying the governing mechanism, choosing a suitable model and checking predictions against the available evidence. I also build Python and solver-API workflows that make simulation studies repeatable and reduce manual work, using AI-assisted development with engineering review of assumptions, settings and results.

PROFESSIONAL EXPERIENCE

3D CFD Simulation Engineer, Robert Bosch, Mexico

2022 to present

- **Proportional hydraulic valve, 1D/3D model calibration.** A single parameter fit could not represent both hot and cold operation. I traced the limitation to changing viscous and inertial loss mechanisms, then developed a regime-split calibration in Python through the Amesim API. The resulting model achieved RMSE at or below 6.13% against 3D CFD across -30 to 120 °C. The calibration isolates parameters in the regimes where they can be identified, producing a model the systems team can deploy without runtime optimisation.
- **Coolant manifold with integrated degassing.** The goal was to remove air without a separate debubbler. I modelled dispersed air transport and evaluated more than six geometry iterations with mechanical designers, balancing bubble residence time and hydraulic losses. The selected design was 3D printed, passed the customer's bench requirements and was integrated into the product platform.
- **Oil cooling inside an electric-motor rotor.** I used VOF, rotating-frame modelling and conjugate heat transfer to assess channel shoulders and feed-pipe alignment. Removing the shoulders increased simulated heat flux by 40% at 5 L/min and 48% at 3 L/min, despite a smaller wetted area. Feed-pipe misalignment changed HTC and heat flux by at most 1.6%, supporting a more manufacturable design.
- **Condensation in a thermal unit containing electronics.** I checked phase-change models against the published CONAN experiment, with a maximum deviation of 4% under its benchmark conditions. A staged solution strategy made longer cooling transients affordable. Comparing sealed and open configurations identified different moisture-accumulation behaviour and informed sealing and component-placement decisions.
- **Multi-port coolant valve, hydraulic optimisation.** I evaluated manufacturable changes against pressure-drop limits and correlated the CFD method with bench data. One adapter redesign reduced simulated path losses from 60 to 28 mbar at 1000 L/h. In a separate split-flow case, removing recirculation pockets reduced losses from 28 to 26 mbar against a 27 mbar target.
- **Compressible throttle flow, virtual characterisation.** I validated the method on a measurable 40 mm valve, with discrepancies of 6–8%, before extending it to larger throttle bodies beyond the available bench capacity. The study covered opening-angle sweeps and resolved compressible flow and shock structures, delivering mass-flow curves for development and calibration.
- **Fuel-cell vehicle, underhood airflow and thermal integration.** I developed the geometry-preparation and meshing workflow for a detailed vehicle model of approximately 100 million cells. Radiators were represented as porous media and fans through performance-based pressure jumps. I delivered radiator airflow across vehicle speeds and fan states, checked front-end pressures against Pitot measurements, and assessed a blocked-radiator scenario to inform cooling-system decisions. The work combined full-vehicle integration with a detailed radiator-section thermal model, linking system-level airflow predictions to the heat-transfer questions that motivated the study.
- **Hydrogen dilution and ventilation.** I used species-transport simulations to assess exhaust-baffle variants and a separate ventilation system, including degraded fan operation. In the baffle study, a geometry change reduced simulated back-pressure from 4612 to 738 Pa at normal flow while retaining outlet hydrogen below 1%. The studies supported design decisions under the modelled operating conditions.

Engineering innovation. Three internal invention disclosures filed; two under review.

SIMULATION AUTOMATION AND TOOLING

- **FlowPilot, automated hydraulic simulation in OpenFOAM.** I built a desktop workflow for design and 1D engineers covering meshing, setup, convergence monitoring and results extraction. A standard four-case study fell from about 180 to 60 minutes; solver time fell from 90 to 25 minutes on 48 CPUs. Pressure-drop predictions were mostly within 10% of the Fluent baseline.
- **Electrolyzer porous transport layer, open-source workflow.** I rebuilt a two-phase workflow in OpenFOAM and compared it with Fluent before relying on it. In the ideal-gas benchmark on 48 CPUs, runtime fell from 82 to 11.2 hours, with a 2% difference in pressure gradient. Relative-permeability results were delivered as a transfer function for the 1D systems team.
- **Pump hydroacoustics and Python post-processing.** Transient pump-pressure analysis required more flexible spectral processing than the solver's standard reports. I developed FFT, power spectral density and band-limited pressure RMS processing for NVH comparisons. The resulting tool became a departmental asset for turbomachinery evaluations.
- **Reusable PyFluent simulation drivers.** Public demonstration studies show configurable meshing, solver setup, operating-point sweeps and post-processing. I use MCP-connected tooling to inspect APIs and develop driver code, while checking model assumptions, boundary assignments and numerical results. The drivers separate configuration from execution so the workflow can be reused and checked when a geometry or operating point changes.
- **Automated spray-layout screening and supplier tolerances.** I applied quasi-steady OpenFOAM screening to nineteen PFI/DI layouts and seventeen ethanol-injection tolerance variants. Weighted criteria combined evaporation, mixture uniformity and wall impingement. The approach identified candidates for a colleague's full-transient validation and converted a separate tolerance study into a usable supplier window.
- **Evaluation of accelerated oil-cooling methods.** I assessed Eulerian and particle-based alternatives using AVL FIRE M, PreonLab and DIVE. Sensitivity to tunable parameters limited predictive confidence, so I recommended against adopting the proposed workflow without further physical validation.

CAPABILITIES

Physics. Conjugate heat transfer; VOF and Eulerian multiphase flow; Lagrangian particles, sprays and wall film; condensation and evaporation; compressible flow; species transport; porous media; rotating machinery with MRF and sliding mesh; fluid-structure interaction.

Tools. Ansys Fluent, OpenFOAM, cfMesh, snappyHexMesh, AVL FIRE M, Simcenter Amesim, PyFluent, Python, MATLAB, Bash, ANSA, Discovery, ParaView, EnSight, Linux and HPC.

Methods. RANS and transient modelling; mesh and time-step sensitivity; benchmark and test comparisons; solver-to-solver checks; design-of-experiments studies; hydraulic optimisation; numerical calibration and solver-API automation.

Languages. Spanish; English.

EDUCATION AND RESEARCH

PhD, Thermofluids Engineering, UNAM 2018 to 2025

Aneurysm haemodynamics: blood rheology, turbulence-model assessment, fluid-structure interaction and particle transport, with benchmark and sensitivity checks.

Master of Engineering, Universidad de Colima 2015 to 2017

Tubular CVD reactor: CFD and experimental analysis, with heat-transfer interpretation using dimensionless groups.

Mechanical and Electrical Engineering, Universidad de Colima 2009 to 2014

Lecturer, Tecnológico de Colima, Mexico 2017 to 2022

PUBLIC TECHNICAL EXAMPLES

Generic demonstrations at albertobraso7.github.io illustrate methods used professionally without reproducing customer components or confidential results.

Pump [/pump](#), degassing [/debubbler](#), throttle flow [/throttle-body](#).

Independent study: residential ventilation [/ventilation](#).

Academic research: aneurysm haemodynamics [/aneurysm](#), CVD reactor [/reactor](#).

PUBLICATIONS

Brambila-Solórzano, A., et al. Influence of blood rheology and turbulence models in the numerical simulation of aneurysms. *Bioengineering* 10, 1170 (2023).

Sánchez, G. M., et al. Numerical simulation of the aqueous humor flow in the eye drainage system. *Med. Eng. Phys.* 83, 82–92 (2020).